APPLIED MULTIVARIATE METHODS
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Homework 05 (50 points)

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eblcat.csv is a dataset containing demographic, preoperative, and intraoperative information for 142 pancreaticresections for pancreatic cancer. It was designed to evaluate relationships between preoperative and intraoperative factors and blood loss during surgery. Below is a brief description of the variables contained in the ebl.csv dataset:

ebl = estimated intraoperative blood loss, expressed in mL
op = type of operation (0= partial resection of the body or tail performed for tumors located in the body or tail,
1= partial resection of the pancreatic head (i.e., "whipple operation"))
ICU = dichotomous variable indicating whether or not the patient required an ICU stay immediately followingsurgery (1=was required to stay, 0=was not required to stay)

We are interested in using logistic regression to examine the association between binary estimatedintraoperative blood loss (predictor) and ICU stay (outcome).

We would like to recode the continuous ebl into binary variable eblcat for use in today's homework. Let'schallenge ourselves with this data management step. Please create a new binary variable "eblcat" that categorizes ebl using the R code:

```
> eblcat$eblcat <- 0
> eblcat$eblcat[eblcat$ebl > 660] <- 1
```

a) Now, fit a model that estimates the chances of staying in the ICU (outcome) based on blood loss (as abinary predictor, eblcat). Write down the estimated model. Include the appropriate output in the answer. (7 points)

$log(\ p/1-p\ )$ = $\beta 0$ + $\beta 1 X$eblcat, $where$, p = $P(Y$ = 1|Xeblcat,)
Y = { 1 if needs ICU 0 if $did\ not$ needs ICU; and 1 if yes eblcat (blood loss)> 660ml, 0 if less than 660ml 2016}

$logit(p$hat) = log ($p/$ 1 − p) = 0.1872 + 0.8142Xeblcat

```
Call:
glm(formula = ICU ~ eblcat, family = "binomial", data = ebldata)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.1872     0.2320   0.807   0.4196
eblcat        0.8142     0.3602   2.260   0.0238 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 186.56  on 141  degrees of freedom
Residual deviance: 181.29  on 140  degrees of freedom
AIC: 185.29
```

Intercept: The value of 0.1872 is the estimated log odds of being in ICU among those who are have eblcat =0 or blood loss less than 660:
- The coefficient of the eblcat variable, the slope of 0.8142, is the log odds ratio for being in ICU if blood loss more than

660 ml or eblcat=1

b) Using the omnibus likelihood ratio test for the logistic regression model, test whether the model in part is meaningful at the 5% level of significance. Include the appropriate output in the answer. (8 points)

The omnibus test for this logistic function is shown below with output. This test will show how at least the one of the model intercept and coefficient is significant at least 5% and 0.02176 is less than 5% which means it is significant.

```
> lrtest(model1)
Likelihood ratio test

Model 1: ICU ~ eblcat
Model 2: ICU ~ 1
  #Df  LogLik Df Chisq Pr(>Chisq)
1   2 -90.647
2   1 -93.280 -1 5.265    0.02176 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

c. From the model in part a), report and interpret the slope of eblcat, relevant odds ratio (OR), and 95% CI. Also, make a conclusion about statistical significance, providing evidence in support. (10 points)

```
> questionr::odds.ratio(model1)
Waiting for profiling to be done...
                 OR   2.5 % 97.5 %      p
(Intercept) 1.20588 0.76633 1.9099 0.4196
eblcat      2.25745 1.12480 4.6406 0.0238 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The odds ratio for eblcat is 2.25745. This OR can be interpreted as the odds of staying in the ICU being about 2.26 times higher for patients with blood loss greater than 660 compared to those with blood loss of 660 or less, holding other variables constant. The OR greater than 1 indicates a positive association between high blood loss and the likelihood of staying in the ICU.
95% Confidence Interval (CI) for OR
The 95% confidence interval for the odds ratio ranges from 1.12480 to 4.6406. This interval does not contain 1, which supports the conclusion that the odds ratio is significantly different from 1, indicating a significant association between eblcat and the odds of staying in the ICU.

d) Fit a model that estimates the chances of staying in the ICU (outcome) based on blood loss (as a binary predictor, eblcat), while adjusting for operation type. Write down the estimated model. Include the appropriate output in the answer. (7 points)

$log\ (\ p/\ 1-p\ ) = \beta 0 + \beta 1 X\text{eblcat}, + \beta 2\text{OP} + where, p = P(Y = 1|X\text{eblcat}, X\text{OP})\ Y = \{\ 1\ if\ yes$ blood loss more than 660 ml or eblcat =1 and 0 if less than 660ml, Xop = { 1 $if\ yes$ whipped operation 0 if tail operation)

$$logit(p\text{hat}\ ) = log\ (\ p/\ 1 - p\ ) = -.2731 + 0.2651 X\text{eblcat} + 1.5981\text{XOP}$$

Intercept: The value of -0.2731 is the estimated log odds of being in ICU among those who are have eblcat =0 or blood loss less than 660 and op = 0 means only tail operation:
- The coefficient of the eblcat variable, the slope of 0.2651, is the log odds ratio for being in ICU if blood loss more than 660 ml or eblcat=1 adjusting for operation type
- The slope of 1.5981, is the log odds ratio for being in OP =1 means whipped operation for adjusting for blood loss more than 660 ml

```
Call:
glm(formula = ICU ~ eblcat + op, family = "binomial", data = ebldata)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.2731     0.2656  -1.028 0.303886
eblcat        0.2651     0.4063   0.653 0.514050
op            1.5981     0.4128   3.871 0.000108 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 186.56  on 141  degrees of freedom
Residual deviance: 164.91  on 139  degrees of freedom
AIC: 170.91
```

e. Using the omnibus likelihood ratio test for the logistic regression model, test whether the model in partis meaningful at the 5% level of significance. Include the appropriate output in the answer. (8 points)

The Omnibus test with 5% significant level shows the p value of 0. The model which includes the Logit function with being admitted in ICU with more than 660 ml blood loss with whipped operation is significant.

```
> lrtest(model2)
Likelihood ratio test

Model 1: ICU ~ eblcat + op
Model 2: ICU ~ 1
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   3 -82.456
2   1 -93.280 -2 21.647  1.992e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
```

F) From the model in part d), report and interpret the slope of eblcat and the slope of op, relevant odds ratios (ORs), and 95% CIs. Also, make a conclusion about statistical significance, providing evidence in support. (10 points)

```
> questionr::odds.ratio(model2)
Waiting for profiling to be done...
                 OR   2.5 %  97.5 %         p
(Intercept) 0.76103 0.44869  1.2777 0.3038859
eblcat      1.30360 0.58202  2.8889 0.5140497
op          4.94347 2.24861 11.4485 0.0001084 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
```

Intercept: The value of -0.2731 is the estimated log odds of being in ICU among those who are have eblcat =0 or blood loss less than 660 and op = 0 means only tail operation:
- The coefficient of the eblcat variable, the slope of 0.2651, is the log odds ratio for being in ICU if blood loss more than 660 ml or eblcat=1 adjusting for operation type
- The slope of 1.5981, is the log odds ratio for being in OP =1 means whipped operation for adjusting for blood loss more than 660 ml

The odds ratio for eblcat is 1.3. This OR can be interpreted as the odds of staying in the ICU being about 1.3 times higher for patients with blood loss greater than 660 compared to those with blood loss of 660 or less, holding OP constant. The P value of 0.514 eblcat suggests that it is not significant. However, the OR greater than 1 indicates a positive association between high blood loss and the likelihood of staying in the ICU.

The odds ratio for OP is 4.9. This OR can be interpreted as the odds of staying in the ICU being about 4.9 times higher for patients with OP 1 which means whipped operation type keeping blood loss constant, holding OP constant. The P value of 0.0001084 eblcat suggests that it is significant. However, the OR greater than 4.9 indicates a very strong relation between being in ICU and having a Whipped pancreatic condition.

95% Confidence Interval (CI) for OR
The 95% confidence interval for the odds ratio ranges for eblcat is from 0.45 to 1.277. This interval contains 1, which supports the conclusion that the odds ratio is significantly different from 1, indicating a potential no or less than association between eblcat and the odds of staying in the ICU.

However, The 95% confidence interval for the odds ratio ranges for OP is from 2.25 to 11.4485. This interval is highly likely that even at the 2.5 percent level of significance it is more than 2.25, and at 97.5 CI level it 11.4485 it mean being an ICU and having OP 1 or whipped pancreatic condition.

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```
setwd("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Multivariate")
library(readr)
rm(list=ls())
ebldata <- read.csv("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Mult/eblcat (3).csv")
ebldata$eblcat <- 0
ebldata$eblcat[ebldata$ebl > 660] <- 1
install.packages("lmtest")
install.packages("car")
install.packages("MASS")
install.packages("lme4")
install.packages("glmnet")
install.packages("lmtest")
library(lmtest)
install.packages("questionr")

model1 <- glm(ICU ~ eblcat, data = ebldata, family = "binomial")
summary(model1)
lrtest(model1)
questionr::odds.ratio(model1)
model2 <- glm(ICU ~ eblcat + op, data = ebldata, family = "binomial")
summary(model2)
lrtest(model2)
questionr::odds.ratio(model2)
lmtest::lrtest(model1, model2)
```